

Skills Summary

Correspondence and Scale Factor in Similar Triangles

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the correspondence

Rule. $\triangle ABC \sim \triangle DEF$ pairs the vertices in the order written: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Matching sides are the ones between matching vertices, so $\overline{AB} \leftrightarrow \overline{DE}$ and $\overline{BC} \leftrightarrow \overline{EF}$.

Write every proportion the same way round — first triangle on top on *both* sides:

$$\frac{AB}{DE} = \frac{BC}{EF}.$$

Example. $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm, $BC = 7$ cm and $DE = 10$ cm. Find EF .

Step 1. Want EF , and I know the side matching it (BC) plus one complete pair (AB with DE) — so one proportion with first triangle on top on both sides does it.

Step 2. $\frac{4}{10} = \frac{7}{EF} \Rightarrow 4 \cdot EF = 7 \cdot 10$
 $EF = 17.5$ cm

Try it. $\triangle ABC \sim \triangle DEF$ with $AB = 5$ cm, $BC = 8$ cm and $DE = 15$ cm. Find EF .

check: $EF = 24$ cm

(!) Watch out: the unknown must sit on the same level as the other side of its own triangle. If $\overline{DE}$ is on the bottom, $\overline{EF}$ is on the bottom too.

Skill 2 — Using the scale factor correctly

Rule. The scale factor *from* one triangle *to* another is

$$k = \frac{\text{length in the target triangle}}{\text{matching length in the starting triangle}},$$

and every side of the starting triangle is multiplied by that same k . Enlarging gives $k > 1$; reducing gives $k < 1$. Reversing the direction replaces k by $1/k$.

Example. $\triangle PQR \sim \triangle STU$ with $PQ = 8$ cm, $ST = 20$ cm and $QR = 6$ cm. Find TU .

Step 1. Want a side of the target triangle, and I have one complete pair to build the factor from — so find k first, then multiply.

Step 2. $k = \frac{20}{8} = 2.5$, so $TU = 6 \cdot 2.5$
 $TU = 15$ cm

Try it. $\triangle PQR \sim \triangle STU$ with $PQ = 9$ mm, $ST = 27$ mm and $QR = 4$ mm. Find TU .

check: $TU = 12$ mm

Skill 3 — Spotting the error in a solution

Two checks that catch almost every setup mistake.

- **Same order twice.** Each ratio must be first triangle over second triangle. If one side of the equation is flipped, the answer comes out by a factor of k^2 .
- **Size sense.** Enlarging must make every length longer, reducing must make every length shorter. An answer that moves the wrong way means the factor was inverted.

Example. $\triangle ABC \sim \triangle DEF$ with $AB = 6$, $BC = 9$ and $DE = 15$. Some work reads $\frac{6}{15} = \frac{EF}{9}$. Redo it correctly.

Step 1. The unknown EF belongs to the second triangle, so it must sit on the bottom like DE does — that tells me the written ratio is upside down before any arithmetic.

Step 2. $\frac{6}{15} = \frac{9}{EF} \Rightarrow 6 \cdot EF = 9 \cdot 15$
 $EF = \mathbf{22.5}$

Try it. $\triangle ABC \sim \triangle DEF$ with $AB = 8$, $BC = 20$ and $DE = 12$. Some work reads $\frac{8}{12} = \frac{EF}{20}$. Redo it correctly.

check: $EF = 13$